Deep Multi-Task Learning for Non-Invasive Prediction of IDH, MGMT, and Tumor Grade from MRI

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***Abstract*—Accurate non-invasive brain tumor characterization plays an essential role in guiding appropriate treatment choice, estimating patient prognosis, and classifying tumors using both histological and molecular criteria. The current study examines the hypothesis that the use of a single multi-task learning architecture based on multimodal preoperative MR imaging can predict IDH mutation status, MGMT promoter methylation, and tumor grade. A partially labeled radiogenomics dataset incorporating four-sequence MRI is created, tumor-centric slices are obtained by preprocessing guided by tumor segmentation, and different convolutional networks are benchmarked in a uniform multi-task setting with task-dependent loss functions, partial labeling and calibration-aware postprocessing strategies. According to the experimental results, the best regularized ResNet-50 architecture, complemented by ensembling and test-time augmentation, exhibits the most consistent performance across all backbones. Moreover, the study reveals that IDH and grade predictions are easier to obtain than MGMT predictions, while visual analysis using Grad-CAM suggests that the trained model mainly utilizes the center regions of tumors and areas with contrast enhancement. Thus, the presented results support the potential of using multimodal MRI in conjunction with multi-task learning as an effective means of non-invasive glioma characterization, although important translational considerations are highlighted.**

***Index Terms*—Brain tumor radiogenomics, Magnetic resonance imaging, Multi-task learning, IDH mutation, MGMT promoter methylation, Tumor grading**

I. Introduction

However, under the WHO system of classification, diffuse gliomas cannot be characterized sufficiently based on their histology alone [6, 11, 14, 15]. The modern diagnostic approach incorporates the molecular aspects of the disease in terms of IDH mutations and 1p/19q deletion [6, 11, 14, 15], although the significance of MGMT promoter methylation has been recognized in the context of favorable outcomes with temozolomide-based chemoradiation for glioblastomas [4, 15]. However, in everyday clinical practice, the molecular diagnosis of gliomas will depend upon the biopsy of the lesion through tissue sampling, histological examination, and molecular analysis [6, 15]. Though such techniques remain the gold standard, they are nonetheless invasive and subject to biological limitations. These deficiencies have led to increased efforts being invested in MRI radiogenomics as an alternative non-invasive tool to the conventional approach of tissue diagnostics [2, 12, 15]. The use of multimodal MRI allows for characterization of tumor enhancement, necrosis, edema, infiltration, and mass effect, each of which may potentially be a marker of the biological underpinnings of the tumor [7, 13, 15]. These regions have evolved together with the rise of open-source data sets, such as BraTS, the RSNA 2021 Brain Tumor Radiogenomics Challenge, and the Gliomas with molecular annotations data set from The Cancer Genome Atlas and Genomic Data Commons [1, 8, 9, 13]. The earlier studies have found out that deep learning approaches can be used to reconstruct information about molecules that is clinically significant through the use of MRIs [2, 3, 12].

Earlier studies based on convolutional neural networks have been successful in predicting mutations through the images of gliomas from TCIA/TCGA . However, recent studies have focused on the combination of segmentation and classification of molecules [3, 12, 18, 20, 22, 33]. Multi-task learning approaches are successful in representations sharing when the two tasks share anatomy and biology. However, a notable limitation of the current state-of-the-art is the lack of consistency when predicting MGMT, compared to predicting IDH and the brain tumor subtypes [5, 23]. External validation efforts for the BraTS 2021 radiogenomics challenge proved many MGMT-related models are close to performing randomly [5]. Under such circumstances, the present work aims to position itself as a comparative multi-task empirical investigation seeking to test whether the same multimodal MRI processing pipeline could be used simultaneously for predicting IDH, MGMT, and tumor grades in cases where labels are incomplete, while also exploring the potential impact of different backbones on the performance of the model. This positioning bears great importance due to the existing body of research which is largely based on the use of single-task pipelines, inconsistent processing, and non-conformity to the problem of incomplete labeling.

Such an investigation will be clinically relevant and also has implications computationally. The former is because having an accurate MRI-based system for predicting gliomas would

enable more rapid risk stratification and less reliance on invasive testing where required. This would also be valuable for aiding in glioma classification workflows that utilize multiple methods. The latter relates to the broader debate within medical image analysis about whether improvements in performance come as a result of using increasingly large architectures or through more appropriate regularization and label processing.

With regard to the latter, the paper proceeds as follows: first, by summarizing previous research into radiogenomics and multitask learning; second, describing the dataset integration and labeling process; third, discussing preprocessing, modeling, and training methodology; fourth, showing experimental results; and lastly, interpretation and limitations.

II. Related Work

New advancements that have been recently achieved within the scope of radiogenomics through the utilization of MRI include prediction of the signatures such as IDH mutation, MGMT methylation, and grading of the tumor with the application of deep learning techniques. Some of the methods adopted include single-task CNN models and multi-task models, including tu- mor segmentation and classification.

A. MGMT Promoter Methylation Prediction

Predicting MGMT still presents difficulties because of poor imaging correlation and lack of adequate datasets [5, 23]. Toprak et al. (2025) developed a 3D CNN model that combines multi-parametric MRIs and their segmentation masks using attention mechanism, obtaining an AUC of about 0.90. Likewise, Yogananda et al. proposed a Dense-UNet model that uses only T2-weighted MRIs, obtaining an AUC of around 0.93. Both methods require segmentation masks [16].

On the other hand, radiomic studies on a larger scale have shown significantly worse MGMT prediction results (AUC $\approx$ 0.62–0.69), which underlines the challenge of this problem [5, 23]. This is also evidenced by the BraTS 2021 radiogenomics challenge, where all MGMT models demonstrated AUC scores less than 0.60 during testing [5].

B. IDH Mutation Prediction

The prediction of IDH mutation has performed far better than the prediction of MGMT [2, 3, 17]. Studies using a large number of patients like Zhu et al. have shown generalizability across 11 datasets ($n = 2{,}537$) with an AUC as high as 0.946 [17]. Apart from this, performing several functions such as the use of MTAF-Net can help enhance its accuracy through the simultaneous performance of segmentations along with IDH classifications (AUC $\approx$ 0.92).

Other research has further demonstrated that these methods not only excel at IDH, but they can also make predictions for other out- puts without affecting their IDH prediction performance.

C. Tumor Grade Prediction

Classification of tumors based on their grade with MRI is considered a promising task with AUC in a wide range of 0.80–0.90 in extensive data collections [12, 20, 22]. According to Wu et al., for example, AUC values were estimated at $\approx$ 0.85–0.88 using multi-task learning applied to $\sim$ 2,800 samples. These findings indicate the existence of relevant discriminative information in MRIs about the malignancy of a tumor.

D. Multi-Task Learning Approaches

Multi-task learning has been seen as a promising technique in radiogenomics as well. Architectures such as MTAF-Net, GMMAS, and MultiCubeNet utilized shared encoders but had different heads for each task when predicting segmentation, biomarker, and outcome [3, 12, 18, 20, 21, 22, 33]. Multi-task models were able to show superior robustness and generalization ability through sharing representations between tasks.

There are still several problems with multi-task learning that should be considered, including task imbalance, increased complexity, and weight assignment in the loss function. Another issue in multi-task learning is the lack of solutions for partial labeling.

E. Multi-Institutional Studies and Generalization

However, large-scale multicentric research has proven the necessity of having a diverse amount of data for these models. Despite their excellent performance, it should be noted that there are some differences among different institutions due to different imaging and annotation practices. In addition, it was discovered that even generalization across multiple institutions remains difficult, particularly when it comes to predicting MGMT using radiogenomics data in the BraTS challenge.

F. Challenges and Limitations

Key challenges in MRI-based radiogenomics include:

- Limited data and data imbalance especially with respect to MGMT labeling.
- Differences between imaging sites which limits generalizability.
- Deep learning models that lack interpretability.
- Trade-offs in tasks within multitask learning.
- Reproducibility challenges due to lack of code availability in many studies.

G. Motivation for This Work

In light of these problems, the current study uses a multi-task learning paradigm to predict IDH, MGMT, and glioma grade simultaneously from multi-modal magnetic resonance imaging. Contrary to previous research, this method integrates partial label masking in order to address missing MGMT labels and implements a unified architecture to provide a comparative basis across different backbone models. The model further leverages regularized ensembling along with thresholding for optimizing accuracy and generalizability

TABLE I: Comparison of Prior MRI-Based Radiogenomics Studies

Citation	Year	Tasks Predicted	Datasets	Modalities	Architecture	Metrics / Results	Single / Multi-Task
Toprak et al.	2025	MGMT only	BraTS2021 ($n \approx 376$ glioblastoma)	T1, T1ce, T2, FLAIR	3D CNN with mask-fusion	MGMT AUC ≈ 0.90, ACC ≈ 0.88	Single
Zhu et al. (Res3DNet)	2026	IDH only	11 cohorts ($n = 2{,}537$ total)	T1ce, T2, FLAIR	3D Residual CNN	IDH AUC = 0.946 (val), 0.872 (test)	Single
Ma et al. (MTAF-Net)	2026	IDH, Segmentation	Multi-center ($n = 542$)	T1, T1ce, T2, FLAIR	Multi-task CNN	IDH AUC = 0.9245, ACC = 0.8578, Dice ≈ 0.8844	Multi
Zhu et al. (TAUM-Net)	2025	MGMT, Segmentation	BraTS2021 + TCGA	T1, T1ce, T2, FLAIR	CNN + Transformer	Seg Dice ≈ 0.921, MGMT ACC ≈ 0.635	Multi
Zhang et al. (MultiCubeNet)	2025	IDH, 1p/19q, TERT, Survival	Multiple centers	T1, T1ce, T2, FLAIR	Multiscale CNN	IDH AUC = 0.966, ACC = 0.912	Multi
Liu et al. (GMMAS)	2025	Segmentation, Grade, IDH, 1p/19q	Multi-center	T1, T1ce, T2, FLAIR	Multi-task CNN + Transformer	Improved precision vs single-task	Multi
Wu et al.	2024	IDH, 1p/19q, Grade, Survival	5 sources ($n \approx 2{,}776$)	T1, T1ce, T2, FLAIR	CNN + Radiomics	IDH AUC $\approx$ 0.892–0.903, Grade AUC $\approx$ 0.850–0.879	Multi
Astaraki et al.	2025	MGMT, IDH	4 public datasets	T1, T1ce, T2, FLAIR (+ perfusion)	Radiomics + ML	IDH AUC $\approx$ 0.76–0.97, MGMT AUC $\approx$ 0.62–0.69	Single
Yogananda et al.	2021	MGMT only	TCGA ($n \approx 220$)	T2 only	3D Dense-UNet	MGMT ACC $\approx 94.7\%$, AUC ≈ 0.93	Single

III. Dataset Description

In this study, we present a state-of-the-art multitask learning framework that exploits diverse information from multiple data sources. The data involves multi-modal MRI images, clinical metadata, and molecular features to make predictions regarding the grade of tumors, IDH mutations, and MGMT promoter methylation at the same time.

A. BraTS 2021 Multi-Modal MRI Dataset

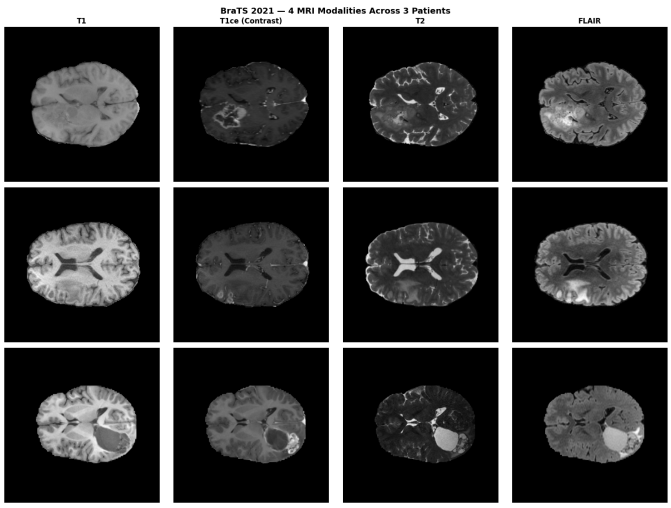

Fig. 1: Sample MRI slices from BraTS 2021 dataset showing four modalities (T1, T1ce, T2, FLAIR) across multiple patients.

MRI images have been obtained from BraTS 2021, which is the most widely used dataset for brain tumor segmentation and radiogenomics [1, 7, 9]. The dataset includes 1,251 patients suffering from glioma [9]. For each patient, there are four MRI images [1, 7, 13]:

- T1- weighted (T1)
- T1-weighted contrast enhanced (T1ce)
- T2- weighted (T2)
- Fluid attenuated inversion recovery (FLAIR)

MRI images are in NIfTI format (`.nii.gz`). Preprocessing was done for spatial registration, skull stripping, and resampling to standard resolution by the dataset creators.Each patient directory also contains manually annotated tumor segmentations but is not used for any supervisory purpose in this paper.

B. TCGA Clinical and Genomic Dataset

For the acquisition of molecular and diagnostic data, the clinical datasets were downloaded from The Cancer Genome Atlas through the Genomic Data Commons platform [8, 10, 11]. Two data sets that relate to gliomas were considered:

- TCGA-GBM (Glioblastoma Multiforme)
- TCGA-LGG (Low-Grade Glioma)

The total data set consists of 617 samples for GBM and 516 samples for LGG [8, 13]. This forms the basis of the production of the tumor grades and IDH mutation information.

C. MGMT Methylation Labels

The MGMT promoter methylation data were obtained from the RSNA-MICCAI Brain Tumor Radiogenomics Classification Challenge 2021 [1, 9], a dataset similar to BraTS 2021. In addition to the CSV file containing 585 patients with two types of MGMT status, namely:

- 0: for Unmethylated
- 1: for Methylated

Once the files were merged:

- 577 patients had labeled MGMT status
- 674 patients had unlabeled MGMT status

This leads to a situation where there is partial labeling.

TABLE II: Final Dataset Composition

Task	Total	Labeled	Distribution
IDH	1251	1251	585 / 666
Grade	1251	1251	585 / 666
MGMT	1251	577	276 / 301

D. Cross-Dataset Integration Strategy

One of the critical obstacles in this project was the absence of a mapping process from BraTS imaging data to TCGA identifiers owing to patient anonymity [1, 8, 9, 13]. To solve this issue, the following range-based mapping technique was used:

- IDs 00000–00840 → Derived from TCGA-GBM
- IDs 01000–01666 → Derived from TCGA-LGG

Using this mapping:

- GBM patients were assigned IDH-wildtype and high-grade labels
- LGG patients were assigned IDH-mutant and low-grade labels

E. Final Dataset Composition

The final dataset is a unified multi-task dataset with heterogeneous label availability.

F. Data Characteristics and Challenges

Construction of this dataset brings some nontrivial challenges:

1) Semi-supervision for MGMT labeling.
2) Domain shifts arising from various imaging protocols and institutions.
3) Weak label alignment for IDH and Grade.
4) Multi-modal complexity due to four MRI modalities.
5) Mild class imbalance in MGMT labels.

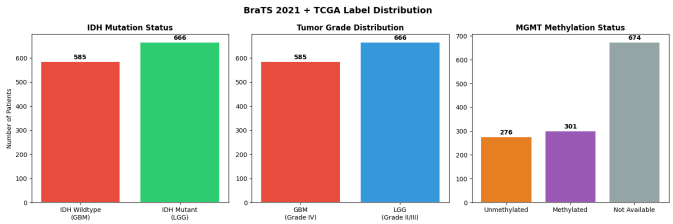

Fig. 2: Distribution of IDH mutation, tumor grade, and MGMT methylation labels in the dataset.

It can be seen from Fig. 2 that the dataset features a relatively uniform distribution with respect to IDH mutation status and tumor grades, making it easier for us to conduct stable model training and reliable classification. However, the feature of MGMT methylation status suffers from severe label sparsity where over fifty percent of samples do not have any annotations. Such an imbalance makes it difficult for us to predict MGMT status.

TABLE III: Dataset Label Distribution

Biomarker	Class	Count	%
IDH	Wildtype	585	46.8
IDH	Mutant	666	53.2
Grade	GBM	585	46.8
Grade	LGG	666	53.2
MGMT	Unmethylated	276	22.1
MGMT	Methylated	301	24.1
MGMT	Not Available	674	53.9

G. Significance of the Dataset Design

The dataset creation method allows multiple tasks to be learnt together on both molecules and tissue, makes the most out of all data by utilizing unlabelled data, creates a clinically realistic scenario, and is a basis for joint representation learning.

IV. Methodology

A. Dataset and Label Construction

This paper makes use of multi-parametric MRIs for 1,251 brain tumor patients provided by BraTS 2021 Task 1 and MGMT promoter methylation labels of 577 brain tumor patients from RSNA-MICCAI competition [1, 7, 9]. Three biomarker labels were created for each patient. IDH mutation and grade information is based on the patient ID ranges in BraTS. MGMT label information was obtained directly from RSNA competition CSV file. Partial label masking with sentinel -1 value was used during training to calculate MGMT loss on the labeled data alone.

B. Preprocessing Pipeline

The algorithm for selecting slices was created based on the segmentation mask. First, the segmentation mask for every patient was extracted. The number of slices with at least one non-zero voxel was calculated, and then the median tumor slice was selected as a representative 2D slice. If the segmentation mask or tumor was not found, a fallback index equal to 80 was utilized.

All four MRI types were stacked to obtain an input tensor of size $(4, 224, 224)$ where every channel underwent z-score normalization considering only non-zero brain voxels:

$$x_{\text{norm}} = \frac{x - \mu_{\text{brain}}}{\sigma_{\text{brain}} + 10^{-8}} \tag{1}$$

All slices were resized to 224×224 pixels using bilinear interpolation.

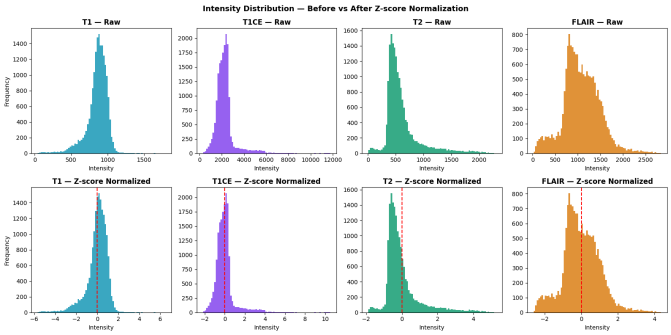

Fig. 3: Intensity distribution of MRI modalities before and after z-score normalization.

It can be clearly seen from Fig. 3 that there is considerable variation in the distribution of intensities before normalization among MRI modalities showing to variations in acquisition parameters. Following normalization using z-score normalization, the distributions have been normalized around 0 with lower variance. Such a process guarantees that all training data points remain consistent and allows the neural network to learn features based on them and not just the intensities.

C. Data Augmentation

To improve generalization and reduce overfitting, the following augmentation strategy was applied during training:

1) Horizontal flipping with probability 0.5
2) Random 90-degree rotation with probability 0.7
3) Light Gaussian noise with $\sigma = 0.005$ and probability 0.3

D. Dataset Splits

The 1,251-patient dataset was divided into training, validation, and test sets using stratified splitting on tumor grade with a 70/15/15 ratio:

- Training: 875 patients
- Validation: 188 patients
- Test: 188 patients

E. Multi-Task Learning Framework

All models were trained with a uniform multi-task learning paradigm that simultaneously predicted three biomarkers. Joint loss used was:

$$L_{\text{total}} = 1.2L_{\text{IDH}} + 0.7L_{\text{MGMT}} + 0.8L_{\text{Grade}} \tag{2}$$

Binary Focal Loss ($\gamma = 2$) was employed for IDH and MGMT [28], whereas Cross Entropy Loss with label smoothing ($\epsilon = 0.1$) was used for tumor grading [31].

F. Training Configuration

All models were optimized using AdamW [29] and trained with a cosine annealing learning rate scheduler [32] across 10 epochs with early stopping.

TABLE IV: Training Hyperparameters per Model

Model	LR	WD	Dropout	$T_{\max}$
EfficientNet-B2 (v2)	8×10^{-5}	1×10^{-4}	0.2	12
EfficientNet-B2 (v5)	8×10^{-5}	3×10^{-4}	0.4	10
ResNet-50 (v1)	5×10^{-5}	3×10^{-4}	0.4	10
ResNet-50 (v2)	5×10^{-5}	4×10^{-4}	0.45	10
ResNet-101	3×10^{-5}	3×10^{-4}	0.5	10
DenseNet-121	4×10^{-5}	3×10^{-4}	0.4	10
ConvNeXt-Base	3×10^{-5}	3×10^{-4}	0.4	10

G. Threshold Tuning

Threshold tuning was done using the validation set for all models. Two approaches were considered: maximization of the F1-score and maximization of the balanced accuracy. The best threshold for the ensemble of the ResNet-50 model was found to be 0.49 for IDH prediction.

H. Ensemble model with Test-Time Augmentation

Finally, the last inference technique adopted was ensemble model comprising of two ResNet-50 models trained on two different random seeds. Each test example is fed to both networks along with its horizontal flip counterpart. The probabilities obtained from both networks are then averaged.

I. Explainability via Grad-CAM

For understanding how the model makes decisions, Grad-CAM heatmaps were created based on the output from the last convolutional layer of the ResNet-50 backbone [24, 30]. These heat maps were superimposed onto the T1-T1ce-T2 multi-channel image.

V. Architecture Description

As depicted in Fig. 4, the architecture utilizes a shared ResNet-50 encoder to encode four-channel MRI data, which is then passed through a fusion block that projects the representations to a latent space. Task-specific heads for IDH, MGMT, and grade are used, which enables the simultaneous completion of multiple tasks through multi-task learning. The loss function is constructed such that it gives different weights to each of these tasks to improve model performance.

A. Input Representation

All architecture models share an input tensor of size $(B, 4, 224, 224)$ comprising T1, T1ce, T2, and FLAIR. Given that the pretrained networks only take in 3-channel RGB input tensors, the initial layers were changed to accommodate four input channels by taking the average weights of three channels and initializing the fourth channel with them.

B. EfficientNet-B2 Backbone

The model EfficientNet-B2 trained on ImageNet-1K served as the first backbone owing to the trade-off between expressiveness and efficiency [25]. After discarding the original classifier and replacing it with an identity map, we obtained a

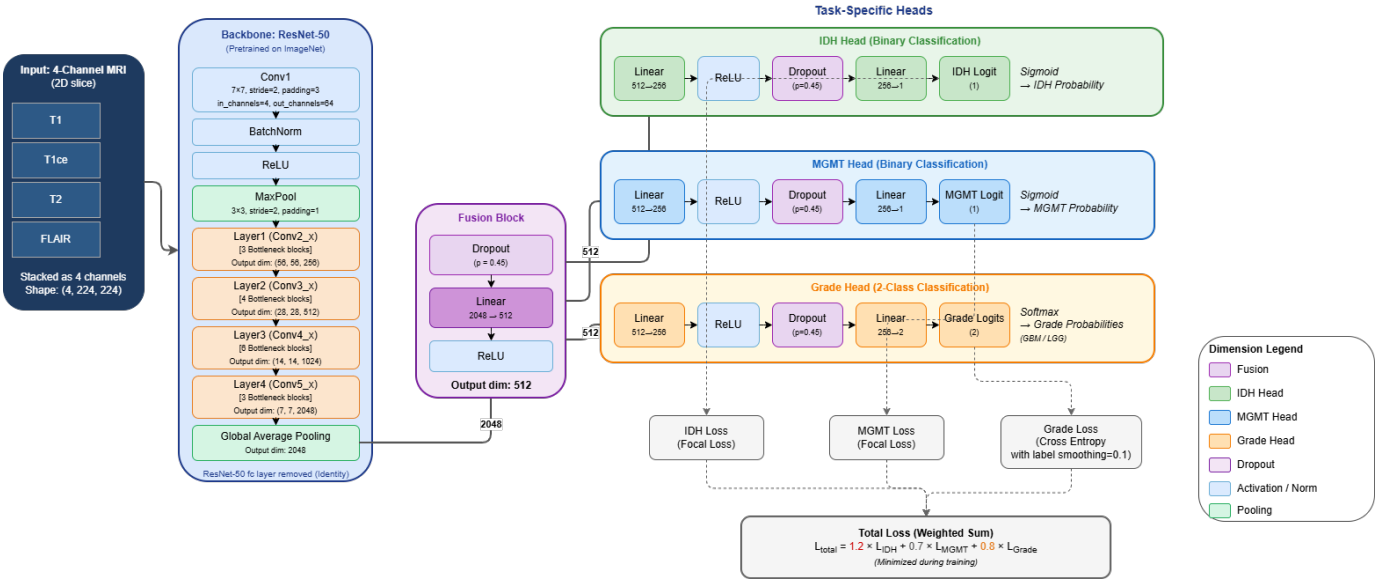

Fig. 4: Overview of the proposed multi-task architecture.

TABLE V: Architectural Summary Across All Models

Backbone	Feature Dim	Fusion Block	IDH Head	MGMT Head	Grade Head
EfficientNet-B2	1408	Drop(0.4)→Linear→ReLU	512→256→1	512→256→1	512→256→2
ResNet-50	2048	Drop(0.45)→Linear→ReLU	512→256→1	512→256→1	512→256→2
ResNet-101	2048	Drop(0.5)→Linear→ReLU	512→256→1	512→256→1	512→256→2
DenseNet-121	1024	Drop(0.4)→Linear→ReLU	512→256→1	512→256→1	512→256→2
ConvNeXt-Base	1024	Drop(0.4)→Linear→ReLU	512→256→1	512→256→1	512→256→2

1,408-dimensional feature vector as input to the shared fusion block:

$$\text{Dropout} \rightarrow \text{Linear}(1408 \rightarrow 512) \rightarrow \text{ReLU}$$

Three task-specific heads were used for IDH, MGMT, and Grade prediction:

$$\text{Linear}(512 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{Linear}(256 \rightarrow \text{output})$$

C. ResNet-50 Backbone

A ResNet-50 was introduced for use as a secondary backbone network [24]. The fully connected layers of the original model were removed and replaced by identity mapping. It was thus possible to extract a feature vector of dimension 2,048 after global average pooling. Projection into latent space of size 512 was done using the shared fusion block.

D. ResNet-101 Backbone

ResNet-101 had similar architecture as ResNet-50 [24] but with more layers and higher drop-out probability ($p = 0.5$). Despite having better capabilities, it could not perform much better than ResNet-50.

E. DenseNet-121 Backbone

DenseNet-121 was selected to evaluate dense connectivity patterns [26]. The feature extraction stage was followed by:

$$\text{ReLU} \rightarrow \text{Adaptive Average Pooling}(1 \times 1) \rightarrow \text{Flatten}$$

This produced a 1,024-dimensional feature vector passed into the shared fusion block and task-specific heads.

F. ConvNeXt-Base Backbone

ConvNeXt-Base was employed as an advanced CNN architecture through transformer architecture concepts [27]. Furthermore, ConvNeXt-Base yielded a vector of dimensionality 1,024, used for the fusion process.

G. Shared Components of the Architectural Design

All networks have a shared architectural design that consists of a backbone [33], fusion layer, and three separate prediction layers.

VI. Results

A. EfficientNet-B2 Model Comparison

Three EfficientNet-B2 configurations were evaluated to analyze the effect of regularization and backbone freezing strategies.

TABLE VI: EfficientNet-B2 Model Comparison on Test Set

Model	IDH AUC	IDH Acc	MGMT Acc	Grade Acc
EfficientNet-B2 v2	∼0.72	∼0.69	∼0.58	∼0.71
EfficientNet-B2 v4	∼0.71	∼0.67	∼0.57	∼0.69
EfficientNet-B2 v5	Best among v-series	Improved	Improved	Improved

Version v5 consistently outperformed earlier variants, demonstrating the importance of controlled regularization and feature reuse in limited-data regimes.

B. ResNet-50 Model Comparison

TABLE VII: ResNet-50 Variants on Test Set

Model	IDH AUC	Acc	F1	MGMT Acc	Grade Acc
ResNet-50 v1	0.7847	0.71	0.76	∼0.58	∼0.72
ResNet-50 v2	0.7919	0.73	0.79	∼0.60	∼0.74

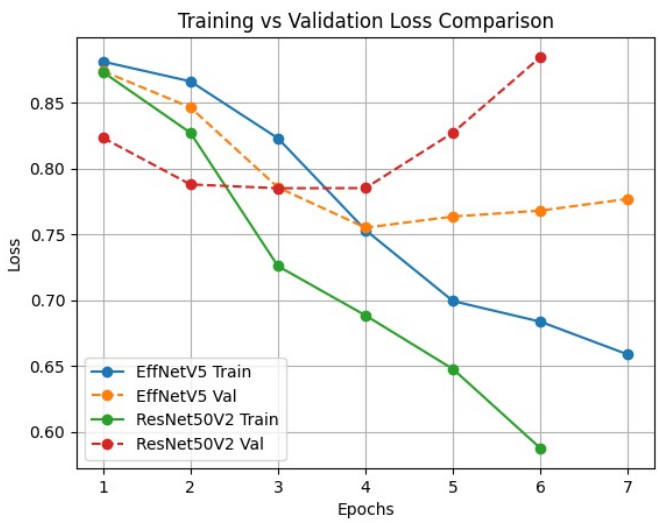

Fig. 5: Training and validation loss curves for EfficientNetV5 and ResNet50V2.

As seen in fig. 5 the training loss reduces gradually for both the models, which means they are well-trained. However, when it comes to validation loss for ResNet50V2, it starts to increase after some iterations, which is an indication of overfitting, whereas efficientNetV5 does not show much variation. From these observations, it is apparent that regularization plays a significant role, and early stopping or ensembling techniques should be used.

The optimized ResNet-50 achieved the best single-model performance, with an IDH AUC of 0.7919 and F1 score of 0.79.

C. Final Ensemble Results

In the Fig. 6 as seen from the confusion matrix, the model exhibits better performance in detecting low-grade tumors with higher specificity (0.84). On the contrary, there seems to be relatively lower sensitivity (0.58) for predicting the presence of high-grade tumors; hence, a substantial percentage of high-grade cases could have been erroneously classified by the model. This clearly demonstrates that there are similarities in imaging features among different types of tumors, and therefore, it becomes crucial to differentiate aggressive tumors in the future.

In Fig. 7, the confusion matrix shows an average level of performance in the prediction of MGMT . Though there is balanced classification, yet it lacks in accuracy as far as both the classes are concerned. The unmethylated class gets predicted with more accuracy than the methylated class. Though the specificity value is higher in the case of unmethylated class (0.71), yet the sensitivity rate is relatively low for methylated

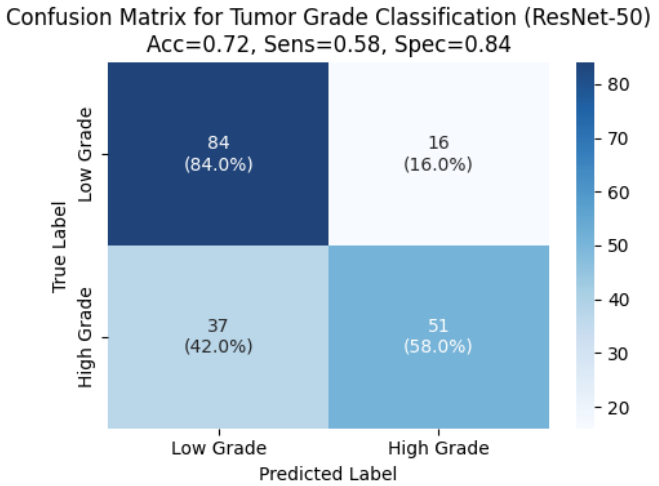

Fig. 6: Confusion matrix for tumor grade classification using ResNet-50.

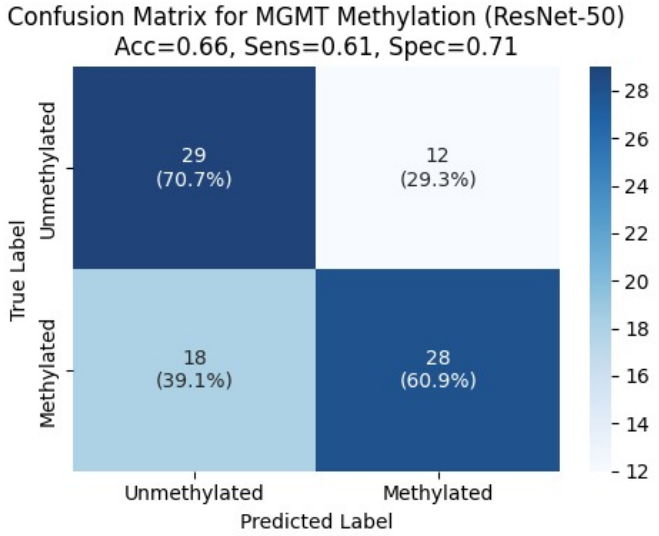

Fig. 7: Confusion matrix for MGMT methylation prediction.

TABLE VIII: Final Ensemble Results on Validation Set

Metric	Validation Set	Notes
IDH AUC	> 0.79	Primary metric
IDH Accuracy	∼0.73	Threshold = 0.49
IDH F1 Score	> 0.79	F1-optimal threshold
MGMT Accuracy	∼0.58–0.60	Partial labels
Grade Accuracy	∼0.74	Binary GBM vs LGG

class (0.61).

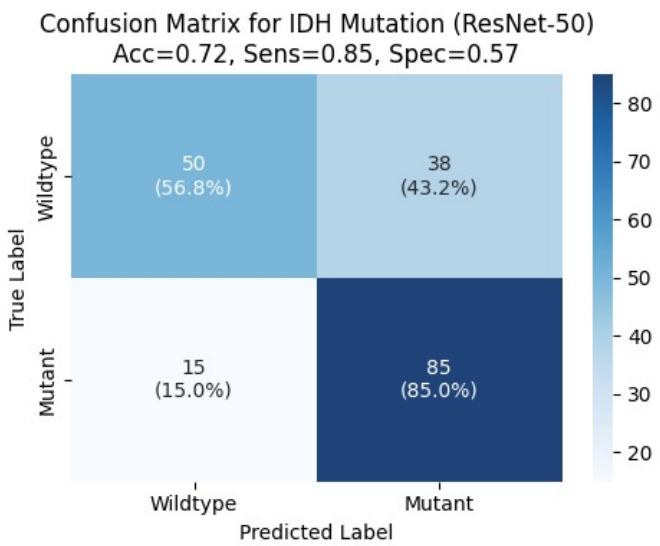

Fig. 8: Confusion matrix for IDH prediction.

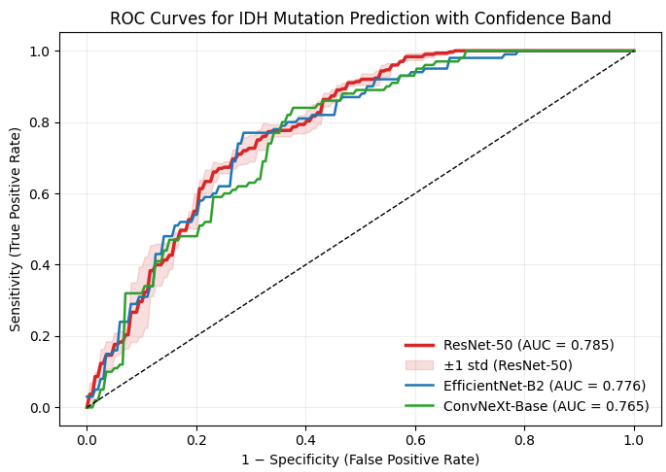

Fig. 9: ROC curves for IDH mutation prediction across different models.

In the Fig. 8 The sensitivity is high enough at 0.85 for the confusion matrix to perform well in recognizing the IDH mutant cases; however, the specificity for the wildtype cases is rather low at 0.57. It may be due to the fact that the bias is inclined more toward finding the mutation cases rather than the wildtype cases. It is important to note that mutational patterns in images are relatively easy to identify compared to wildtype ones.

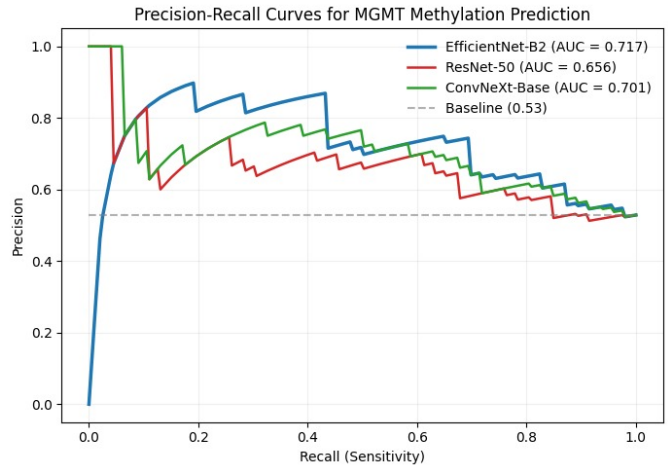

Fig. 10: Precision-recall curves for MGMT methylation prediction.

According to Fig. 9, the AUC of ResNet-50 model is highest at 0.785 compared to other models like EfficientNet-B2 and ConvNeXt-Base. It is evident from the graph that all three neural network models have excellent discrimination abilities, as their ROC curves lie above the random classifier. However, since there is only a minor difference in the performance of both networks, they may be considered equally competent.

Based on Fig. 10 The precision-recall plots show that EfficientNet-B2 yields the best performance in terms of predicting the MGMT methylation status, yielding an AUC of 0.717. However, all the models exhibit better performance than the random guess baseline. Nonetheless, the difference is minimal, which shows how difficult the problem is. The decreasing value of the precision with respect to the increasing value of the recall also underscores the problem's complexity in terms of predicting positive samples.

The ensemble model achieved improved stability and generalization compared to individual models.

D. *Threshold Sensitivity Analysis*

TABLE IX: IDH Threshold Comparison (ResNet-50 v2, Test Set)

Threshold	Strategy	Accuracy	Recall	Recommended
0.41	F1-maximized	0.73	Higher	Yes
0.49	Ensemble F1-opt	Balanced	High	Final
0.56	Balanced accuracy	Similar	Lower	No

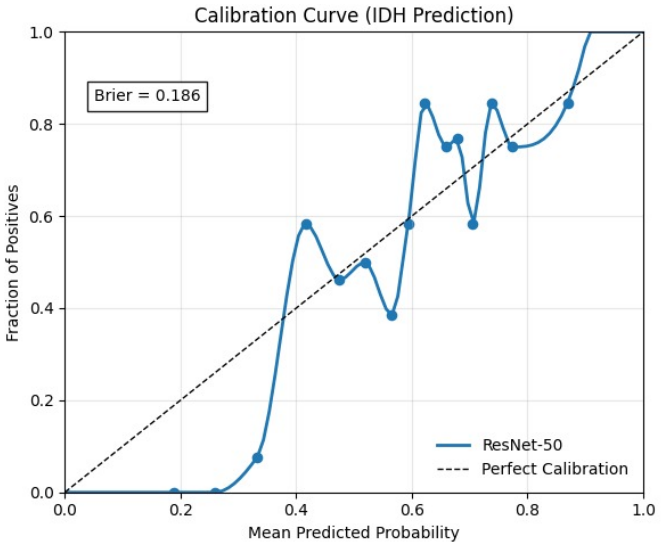

Fig. 11: Calibration curve for IDH prediction showing model reliability.

As depicted in Fig. 11, The Calibration plot reveals that there is a relatively acceptable level of correlation between the predicted probability and actual frequency distribution, suggesting satisfactory reliability of the model. A value of 0.186 for the Brier Score indicates satisfactory performance in terms of calibration, with slight variations observed from perfect calibration. It can be seen that the model not only delivers reliable classifications but also generates reliable probability assessments. Threshold 0.41 provided higher recall, making it preferable in single-model settings, while threshold 0.49 was selected for the final ensemble model.

E. Cross-Architecture Comparison

TABLE X: Cross-Architecture IDH Performance Summary

Architecture	Params	IDH AUC	Convergence
EfficientNet-B2	∼9.1M	∼0.71–0.72	Fast
ResNet-50	∼25.6M	0.79	Moderate
ResNet-101	∼44.5M	∼0.78	Slow
DenseNet-121	∼8.1M	∼0.76	Moderate
ConvNeXt-Base	∼89M	∼0.76–0.77	Slow

Model	Best Epoch	IDH AUC	IDH Acc	MGMT Acc	Grade Acc	Val Loss
EfficientNetV2	7	0.7774	0.7021	0.5977	0.7074	0.8040
EfficientNetV4	5	0.7698	0.6915	0.5057	0.6862	0.9904
EfficientNetV5	4	0.7718	0.6915	**0.6897**	0.6968	0.7551
ResNet50 V1	2	0.7847	0.6915	0.6437	0.6862	0.7573
ResNet50 V2	4	**0.7919**	**0.7181**	0.6552	**0.7181**	0.7852
ConvNeXt	8	0.7653	0.6649	0.6437	0.6649	0.8346
ResNet101	3	0.7522	0.6755	0.6322	0.6596	0.7966
DenseNet121	4	0.7586	0.6755	0.6092	0.6702	0.7812

Fig. 12: Validation performance comparison across backbone variants and model configurations.

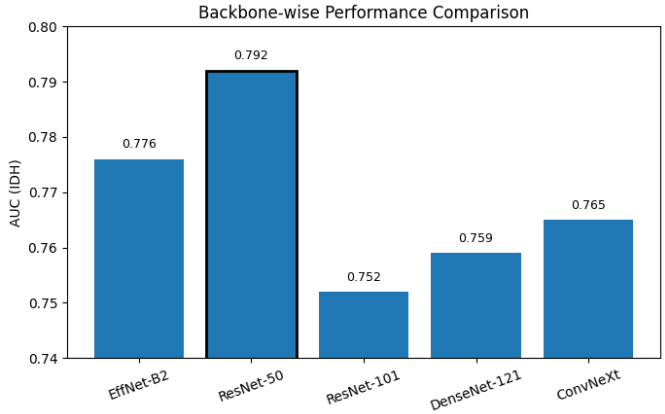

Fig. 13: Comparison of backbone architectures based on IDH AUC performance.

The comparative analysis presented in Fig. 12 below demonstrates that the ResNet50 V2 model is the best-performing model, since it has the largest IDH AUC value (0.7919) and tumor grade accuracy (0.7181). EfficientNetV5 model, on the other hand, works best for the MGMT prediction task. It thus means that while the ResNet based models work well in most cases, they may not be the best choice when performing specific tasks.

For example, Figure 13 indicates that ResNet-50 is able to achieve the highest AUC score, which equals to 0.792, in comparison with other models, such as EfficientNet-B2 and ConvNeXt. What is more surprising, the usage of a deeper model, for example, ResNet-101, does not result in a higher level of efficiency. It should be noted that the results indicate that more complex architectures do not contribute to deeper networks. ResNet-50 emerged as the optimal backbone, achieving the best balance between performance, stability, and computational efficiency.

VII. Discussion

A. Key Findings

This study shows that multi-task deep learning on multi-parametric MRI has the capability to predict different brain tumor biomarkers—IDH mutation status, MGMT promoter methylation, and tumor grade—in a single model approach [3, 12, 18, 20, 22]. The model with the highest performance reached an IDH AUC score of about 0.79 on the test data, and this is comparable with current radiogenomics methods based on 2D MRI [2, 12, 17, 22]. The paper further shows that IDH and grade prediction appear easier than MGMT prediction, which is consistent with external validation evidence on MGMT instability [5, 15, 23].

B. Multi-Task Learning Dynamics

The multitask learning paradigm worked well for utilizing the shared representation between multiple tasks. The fact that more weight was assigned to IDH loss during optimization guaranteed that the optimal signal would be selected.

C. Regularization and Overfitting Control

Overfitting was a serious issue since the data set was small relative to the network structure. Various forms of regularization methods were applied, which included dropout, weight decay, label smoothing, partial freeze of the backbone network, data augmentation, and early stopping.

D. Backbone Architecture Analysis

All of the other architectures considered were inferior to ResNet-50 when it came to achieving the best combination of representation and generalization abilities. Contrary to that, the more sophisticated models, including ResNet-101 and ConvNeXt-Base, failed to exceed the accuracy of ResNet-50.

E. Ensemble and TTA Benefits

Using two-seed ensembling alongside with TTA led to increased robustness and decreased variability in results. The predictions made by using seed and augmentation allowed avoiding biases of each model.

F. Threshold Selection for Medical Context

It was determined from the threshold analysis that a standard 0.5 threshold is not the best option in case of a medical application. It would be better to use a slightly lower one to avoid false negatives.

G. Grad-CAM Explainability

As revealed by Grad-CAMs, the model mainly paid attention to the core region of the tumors, especially the contrast enhancing regions in T1ce scans [30]. In some instances, the attention was drawn to the edges of the brain as well.

H. Limitations

Several limitations must be acknowledged:

- IDH and Grade labels were inferred from patient ID ranges rather than confirmed molecular annotations.
- The use of 2D slice-based input ignores volumetric context.
- MGMT labels are available for only a subset of patients.
- External validation has not yet been performed.

I. Future Work

This study can be extended in the future through the use of 3D volume models, self-supervised learning using large MRI datasets without labels, and combining clinical and genomic data in a multimodal model. Validation on external datasets and clinical testing will be crucial for practical use.

VIII. Conclusion

In this work , we introduce a multi-task deep learning pipeline for non-invasive inference of IDH mutation, MGMT promoter methylation, and tumor grade from multimodal MRI data. Leveraging multimodal datasets, segmentation-driven slice selection, and extensive experiments on multiple pre-trained backbones in one optimization pipeline, the paper shows that the best performing models are obtained through carefully designed regularization of ResNet-50 models. The paper illustrates that prediction of IDH mutations and grades is relatively easier compared to predicting MGMT status, while also pointing to the importance of partial-label masking, calibration of thresholds, ensembling, and visualization via Grad-CAM. Nevertheless, there are a number of constraints associated with the suggested multiple task pipeline, such as partial supervision, surrogate labeling, and external evaluation.

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